Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (EC) Examination

November 2014

EC 201.01/EC 201 Digital Electronics & Logic Design

Date: 12.11.2014, Wednesday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below. **[07]**

a. Which of the following logics is the fastest?

(a) TTL (c) CMOS

(b) ECL (d) LSI

b. The decimal equivalent of 10111.0110 is …………

(a) 22.3 (c) 22.75

(b) 23.75 (d) 23.3

c. Simplify the Boolean function using K-map. $F(x,y,z)=\sum(0,2,4,5,6)$

d. Design a half-adder circuit with a decoder and two OR gates.

e. Perform following operation using 2's complement method (– 46) – (25) = ?

f. Mention name of four different methods for control organization.

g. Define Fan out and Power Dissipation with respect to logic gates.

Q – 2.a Reduce following function using Tabulation Method **[04]**

$$F(w,x,y,z)=\sum(1,4,6,7,8,9,10,11,15)$$

Q – 2.b Answer any two questions. **[10]**

(i) Implement the given function using Multiplexer.

$$F(a,b,c,d)= \sum(0,1,3,4,8,9,15)$$

(ii) Write a short note memory unit and explain read and write functions.

(iii) Explain 4 bit Magnitude Comparator with logic diagram and functions.

Q - 3 Answer any two questions. **[14]**

a. Simplify following function using K-map and draw logic diagram for reduced function

$$F=A'B'C'+B'CD'+A'BCD'+AB'C'$$

b. Design BCD to decimal decoder with Truth table for 0 to 9 decimal numbers.

c. Design BCD to Excess-3 code converter with Truth table for 0 to 9 decimal numbers.

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SECTION – II

Q - 4	**Answer the question below.**	**[07]**
a.	State the difference between combinational and sequential circuit.	
b.	What is race around condition?	
c.	Differentiate between Latch and Flip-flop?	
d.	What is master-slave JK flip flop?	
e.	Write down the characteristic equation of JK flip flop.	
f.	What is the difference between Positive Edge and Positive Trigger clock?	
g.	State the difference between ripple counter and synchronous counter?	
Q – 5.a	Explain the 4 bit Serial input and serial output shift register with proper waveform and diagram.	**[05]**
	OR	
Q – 5.a	Explain static-0 and static-1 Hazard with suitable example.	**[05]**
Q – 5.b	Explain block diagram of Processor unit with scratchpad memory.	**[04]**
Q – 5.c	Explain Positive edge triggered JK flip flop with truth table and Waveform.	**[05]**
	OR	
Q – 5.c	Explain JK flip flop to SR flip flop conversion.	**[05]**
Q – 6.a	What is the Programmable Logic Unit? State the Application of it.	**[02]**
Q – 6.b	Design a Mod 5 Synchronous counter using T flip flop.	**[07]**
Q – 6.c	Design a counter with following binary sequence: - 0,1,3,7,6,4 and repeat.	**[05]**
	OR	
Q – 6.c	Explain Johnson Counter.	**[05]**

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